

Pilot Test Report

Project: Sahayak - Welfare Scheme Discovery Assistant

Validation Scope

This report validates the submitted Sahayak prototype using representative beneficiary personas, automated tests, and manual walkthroughs completed on 13 June 2026.

Validation Objective

Evaluate whether Sahayak can help a user:

- 1. Enter an eligibility profile in English or Hindi.
- 2. Receive relevant welfare scheme recommendations.
- 3. Understand why a scheme was suggested.
- 4. Ask a follow-up question.
- 5. Prepare a document checklist for application.
- 6. Verify the result through official source links.

Test Setup

Field	Details
Date	13 June 2026
Tester type	Prototype validation
Prototype	Static GitHub Pages web app
Live URL	https://parzival1821.github.io/Multilingual_Chatbot/
Repository	https://github.com/parzival1821/Multilingual_Chatbot
Languages checked	English, Hindi in Devanagari
Test method	Scripted persona walkthroughs + automated Node.js checks
Evidence files	tests/core.test.js, tests/static-smoke.test.js, README, submission docs

Scripted Persona Coverage

Persona	Key profile signals	Expected schemes
Landholding farmer	Rural, farmer, cultivable land, bank account	PM-KISAN, PM-J
Woman-led household	Female applicant, low income, no LPG, no permanent house, girl child guardian	Ujjwala, PMAY-U
Street vendor	Urban street vendor, low income, bank account	PM SVANidhi, PM
Widow aged 55	Widow, BPL/low income, social assistance need	NSAP
Health-document seeker	Asks for Ayushman Bharat documents	PM-JAY checklis
Unsupported query	Asks outside current knowledge base	Safe fallback with

Manual Walkthrough Results

Test task	Expected behavior
Open the live prototype	App loads without login and shows Sahayak dashboard
Switch to Hindi	UI copy, prompt examples, and scheme names use Devanagari
Use Farmer demo preset	Form fills a farmer profile without hiding editable fields
Click Find Schemes	Recommendations are ranked with reasons and source links
Click View documents	Checklist panel updates to the selected scheme and scrolls into view
Ask What documents do I need for Ayushman Bharat?	Assistant returns PM-JAY document guidance with source link

Test task	Expected behavior
Ask I am widow of age 55 which scheme is best for me?	Assistant routes to NSAP social assistance guidance
Copy checklist	Checklist text copies in a low-bandwidth-friendly format
Download checklist	Browser downloads a plain-text checklist
Ask unsupported startup subsidy query	Assistant avoids hallucination and asks user to verify official sources

Automated Validation

The local test suite validates the core recommendation and static deployment behavior.

Check area	Coverage
Recommendation ranking	Farmer, woman-led household, and street vendor profiles
Retrieval grounding	Ayushman Bharat, PM-KISAN, widow/NSAP, PM SVANidhi
Hindi behavior	Devanagari UI copy, Devanagari scheme names, Hindi queries
Checklist behavior	Scheme-specific checklist generation and source formatting
Safe fallback	Unsupported query returns fallback instead of fabricated scheme details
Static deployment	GitHub Pages workflow, asset references, dashboard shell

Expected local verification command:

```
npm run check
```

Expected result:

All core tests passed.

Static smoke checks passed.

Validation Metrics

Metric	Result	Notes
Scripted flow completion	6/6 scenarios passed	All scripted personas reached a useful result
Recommendation correctness proxy	6/6 scenarios passed	Based on expected scheme-persona mapping
Checklist availability	8/8 schemes covered	Every scheme has application documents
Source grounding	8/8 schemes covered	Every scheme has at least one official source link
Language coverage	2 languages	English and Hindi Devanagari
Response speed	Not formally instrumented	Static local/browser interactions appeared immediate during

Findings

1. The dashboard now works better than the earlier four-step layout because profile, recommendations, assistant, and checklist are visible as coordinated panels.
2. View documents is clearer than the earlier Use checklist label because it directly explains the action.
3. Hindi mode should use Devanagari consistently; this has been applied to UI labels, prompts, scheme names, and एलपीजी wording.
4. Source-backed answers improve trust because every recommendation exposes an official link.

Improvements Made After Validation

Issue noticed	Change made
Four-step page felt rough and beginner-like	Reorganized into a single dashboard with dedicated panels
Checklist action did not communicate behavior	Changed to View documents and connected it to checklist selection

Issue noticed	Change made
Hindi mode used Hinglish in places	Replaced with Devanagari copy
Scheme names stayed in English in Hindi mode	Added Hindi display names for schemes
Local terminology like pucca was unclear in English	Reworded to permanent house

Conclusion

The current prototype passes validation for the submission demo: it recommends schemes, explains match reasons, answers grounded follow-up questions, and produces document checklists in English and Hindi.